

Brachial Plexus

Department of anatomy

Musculoskeletal System

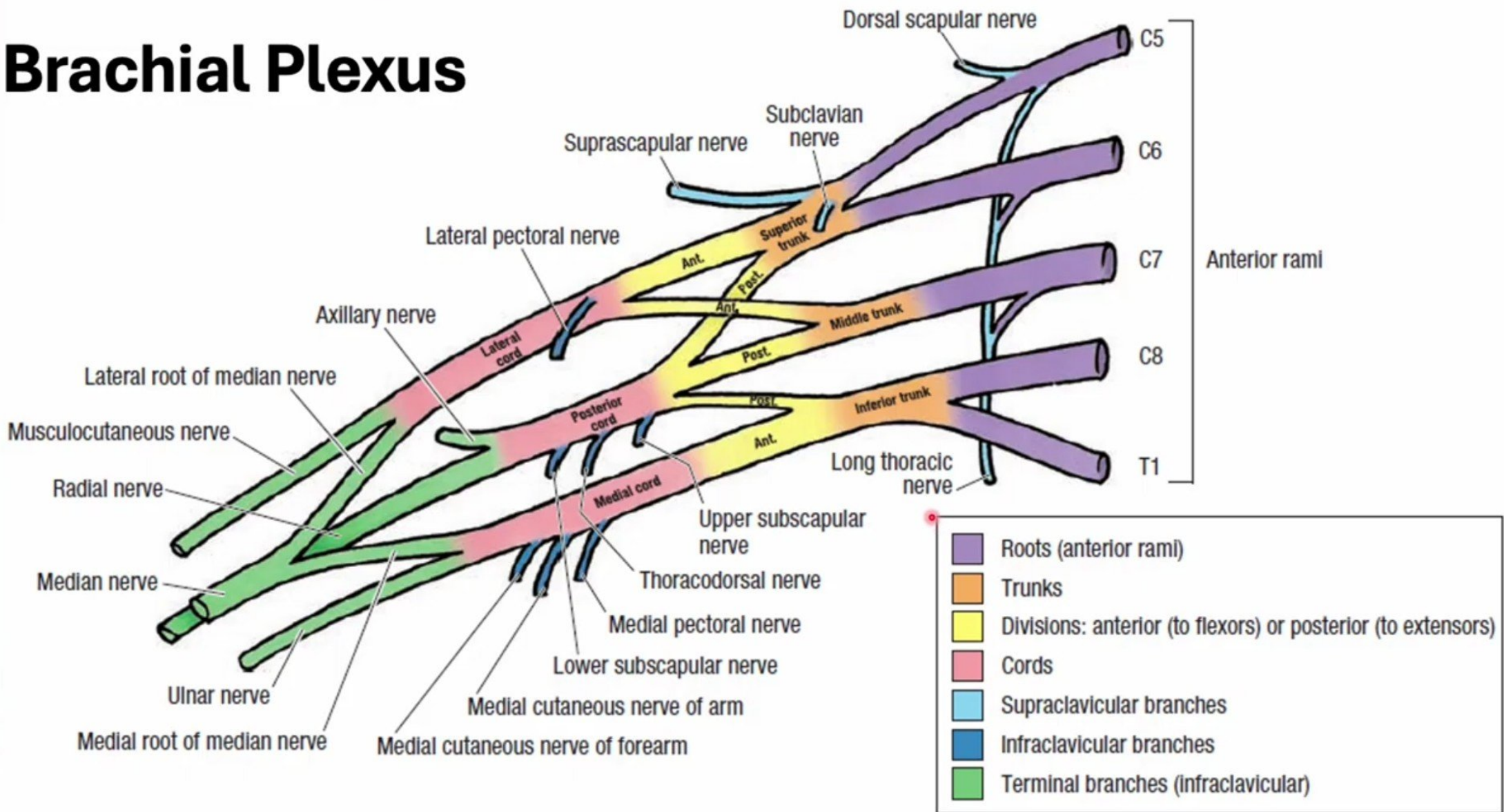
The Upper Limb (4)

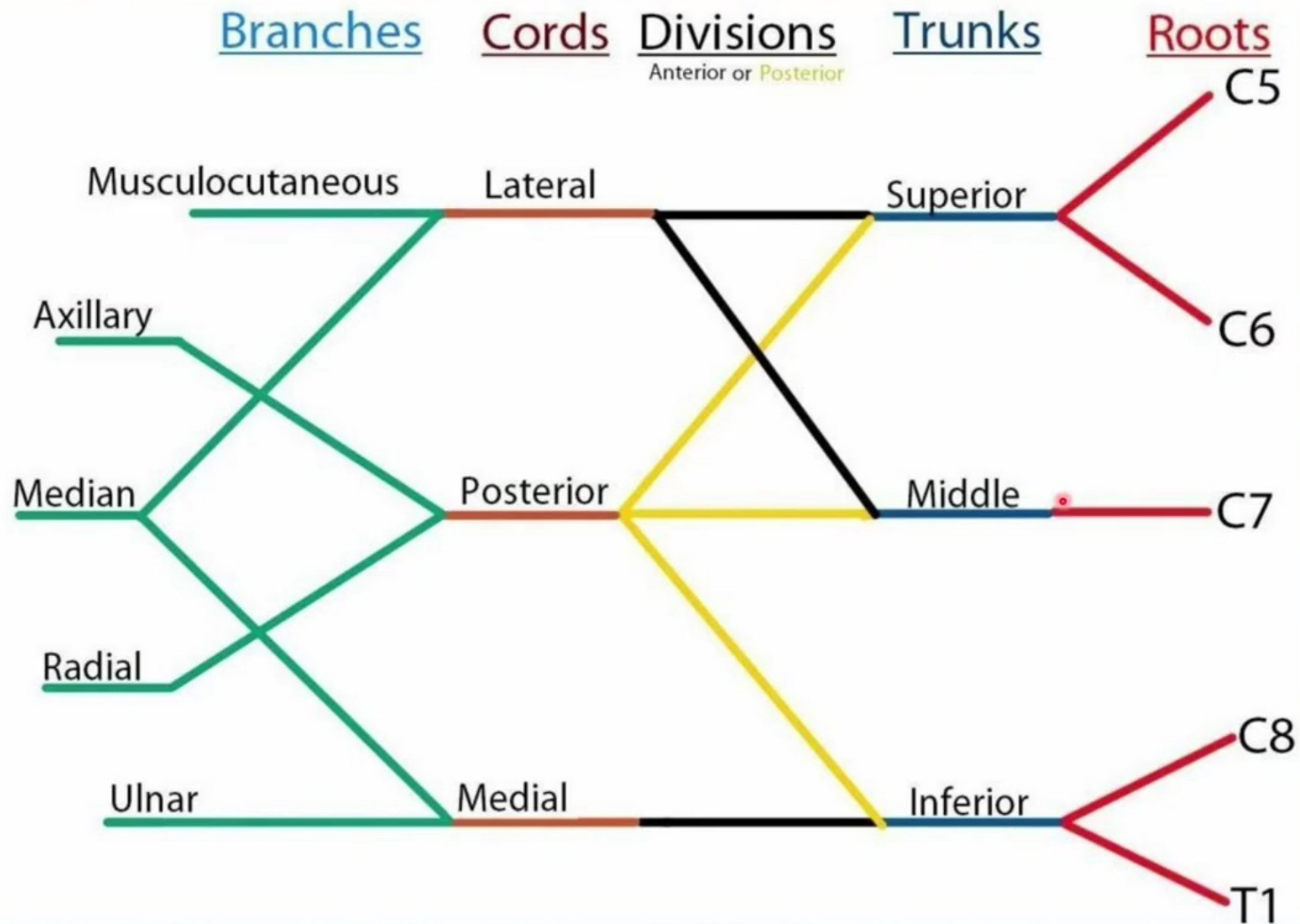
Dr. Osman Ahmed Osman

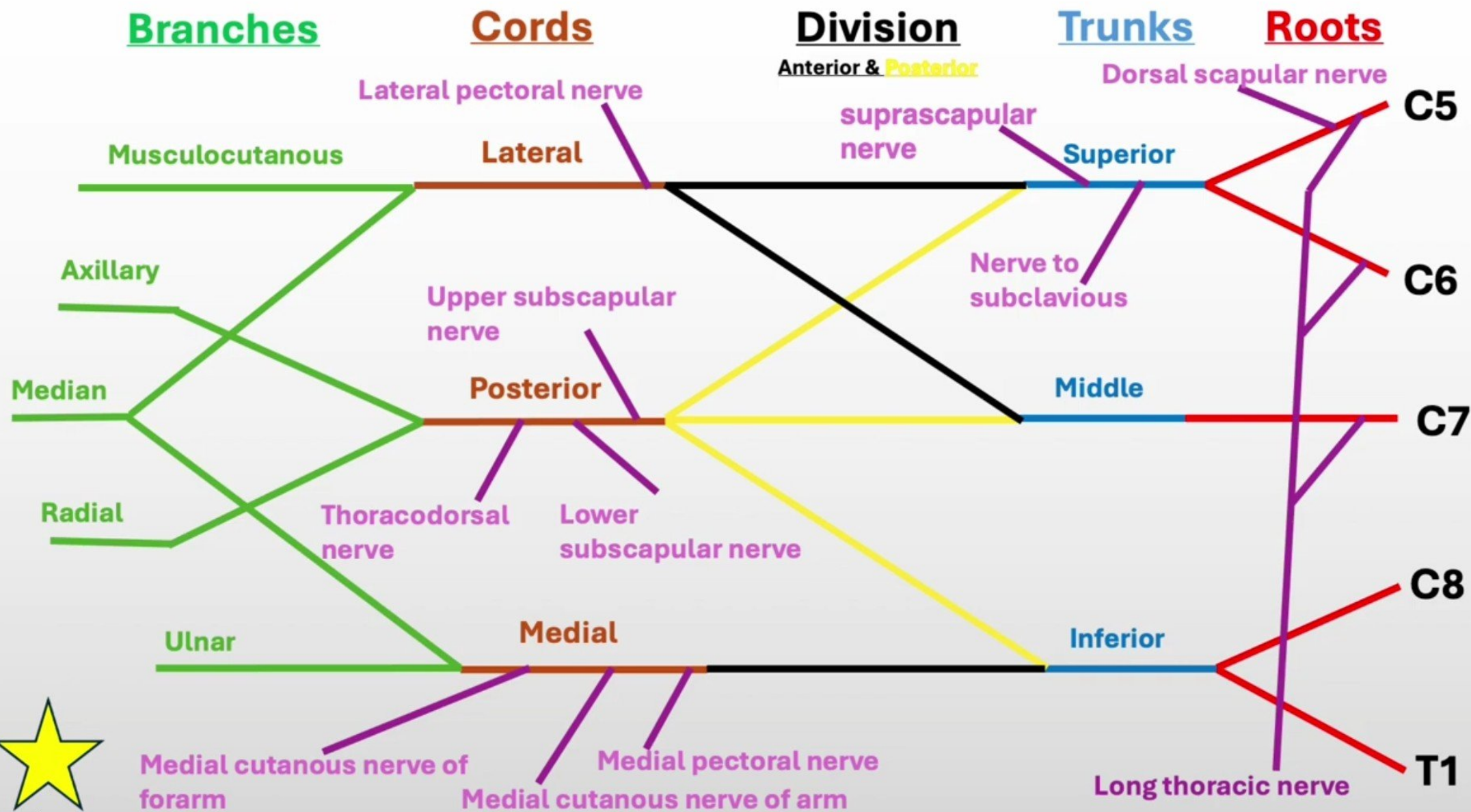
Definition

The brachial plexus (plexus brachialis) is a somatic nerve plexus formed by **intercommunications** among the **ventral rami (Anterior Rami)** of the **lower 4 cervical nerves (C5 to C8)** and the **first thoracic nerve (T1)**.

Brachial Plexus



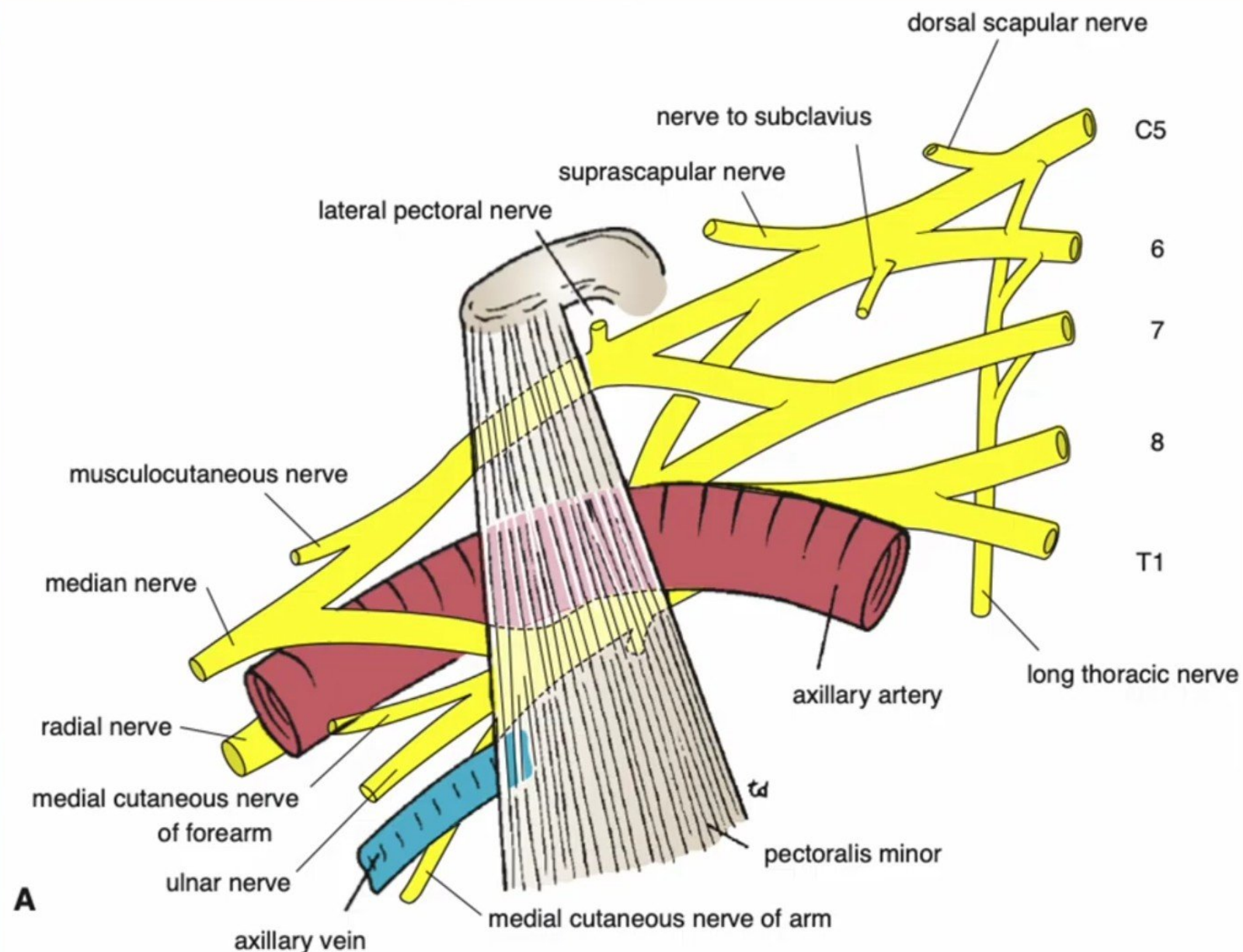




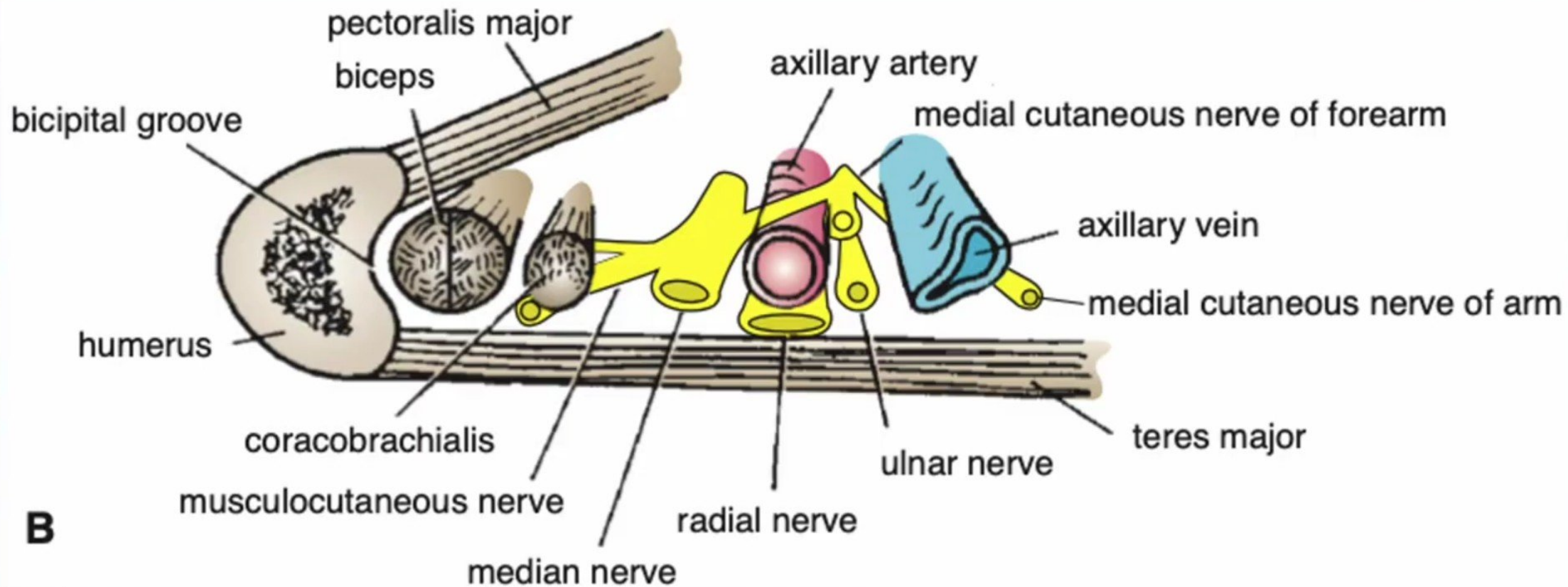
Relations of the brachial plexus and its branches to the axillary artery and vein

➤ **Cords** of the Brachial Plexus All three cords of the brachial plexus lie above and lateral to the first and second parts of the axillary artery.

➤ Most **branches** of the **cords** that form the main nerve trunks of the upper limb continue this relationship to **the third part** of the axillary artery.

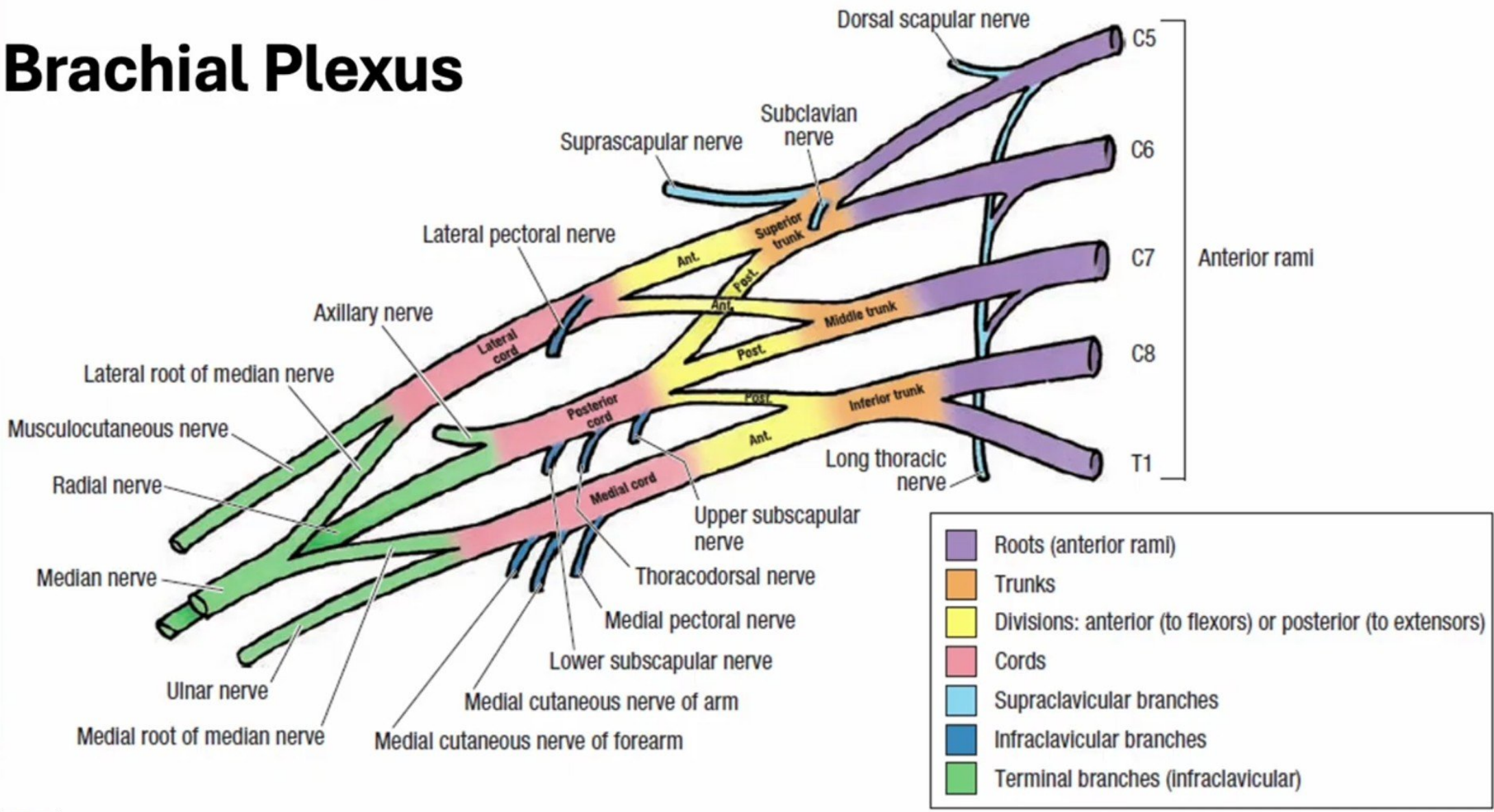


Section through the axilla at the level of the teres major muscle



B

Brachial Plexus



The branches of the brachial plexus

The branches of the different parts of the brachial plexus are as follows:

■ Roots

- ✓ Dorsal scapular nerve (C5).
- ✓ Long thoracic nerve (C5, 6, and 7).

■ Upper trunk

- ✓ Nerve to subclavius (C5 and 6).
- ✓ Suprascapular nerve (supplies the supraspinatus and infraspinatus muscles).

■ Lateral cord

- ✓ Lateral pectoral nerve.
- ✓ Musculocutaneous nerve.
- ✓ Lateral root of median nerve.

■ Medial cord

- ✓ Medial pectoral nerve.
- ✓ Medial cutaneous nerve of arm.
- ✓ Medial cutaneous nerve of forearm.
- ✓ Ulnar nerve.
- ✓ Medial root of median nerve.

■ Posterior cord

- ✓ Upper subscapular nerve.
- ✓ lower subscapular nerve.
- ✓ Thoracodorsal nerve.
- ✓ Axillary nerve.
- ✓ Radial nerve.

Summary of the Branches of the Brachial Plexus and their Distribution

Branches	Distribution
Roots	
Dorsal scapular nerve (C5)	Rhomboid minor, rhomboid major, levator scapulae muscles
Long thoracic nerve (C5, 6, 7)	Serratus anterior muscle
Upper Trunk	
Suprascapular nerve (C5, 6)	Supraspinatus and infraspinatus muscles
Nerve to subclavius (C5, 6)	Subclavius
Lateral Cord	
Lateral pectoral nerve (C5, 6, 7)	Pectoralis major muscle
Musculocutaneous nerve (C5, 6, 7)	Coracobrachialis, biceps brachii, brachialis muscles; supplies skin along lateral border of forearm when it becomes the lateral cutaneous nerve of forearm.
Lateral root of median nerve (C5, 6, 7)	See medial root of median nerve

Summary of the Branches of the Brachial Plexus and their Distribution

Branches	Distribution
Posterior Cord	
Upper subscapular nerve (C5, 6)	Subscapularis muscle
Thoracodorsal nerve (C6, 7, 8)	Latissimus dorsi muscle
Lower subscapular nerve (C5, 6)	Subscapularis and teres major muscles
Axillary nerve (C5, 6)	Deltoid and teres minor muscles; upper lateral cutaneous nerve of arm supplies skin over lower half of deltoid muscle
Radial nerve (C5, 6, 7, 8; T1)	Triceps, anconeus, part of brachialis, extensor carpi radialis longus; via deep radial nerve branch supplies extensor muscles of forearm: supinator, extensor carpi radialis brevis, extensor carpi ulnaris, extensor digitorum, extensor digiti minimi, extensor indicis, abductor pollicis longus, extensor pollicis longus, extensor pollicis brevis; skin, lower lateral cutaneous nerve of arm, posterior cutaneous nerve of arm, and posterior cutaneous nerve of forearm; skin on lateral side of dorsum of hand and dorsal surface of lateral three and a half fingers; articular branches to elbow, wrist, and hand.

Summary of the Branches of the Brachial Plexus and their Distribution

Branches	Distribution
Medial Cord	
Medial pectoral nerve (C8; T1)	Pectoralis major and minor muscles
Medial cutaneous nerve of arm joined by intercostal brachial nerve from second intercostal nerve (C8; T1, 2)	Skin of medial side of arm
Medial cutaneous nerve of forearm (C8; T1)	Skin of medial side of forearm
Ulnar nerve (C8; T1)	Flexor carpi ulnaris and medial half of flexor digitorum profundus, flexor digiti minimi, opponens digiti minimi, abductor digiti minimi, adductor pollicis, third and fourth lumbricals, interossei, palmaris brevis, skin of medial half of dorsum of hand and palm, skin of palmar and dorsal surfaces of medial one and a half fingers.
Medial root of median nerve (with lateral root) forms median nerve (C5, 6, 7, 8; T1).	Pronator teres, flexor carpi radialis, palmaris longus, flexor digitorum superficialis, abductor pollicis brevis, flexor pollicis brevis, opponens pollicis, first two lumbricals (by way of anterior interosseous branch), flexor pollicis longus, flexor digitorum profundus (lateral half), pronator quadratus; palmar cutaneous branch to lateral half of palm and digital branches to palmar surface of lateral three and a half fingers; articular branches to elbow, wrist, and carpal joints.

Lesions of the brachial plexus

Knowledge of the muscles innervated by branches of the brachial plexus and the actions of these muscles and areas of anesthesia and/or paraesthesia allows the physician to determine the localization of a given lesion.

Thanks